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Combining gait analysis and finite element modeling to optimize offloading insoles for calcaneal ulcers

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Running Head: A COMBINED ANALYSIS FOR CALCANEAL ULCERS

Complete Manuscript Title: COMBINING GAIT ANALYSIS AND FINITE ELEMENT MODELING TO OPTIMIZE OFFLOADING INSOLES FOR CALCANEAL ULCERS

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COMBINING GAIT ANALYSIS AND FINITE ELEMENT MODELING TO OPTIMIZE OFFLOADING INSOLES FOR CALCANEAL ULCERS

Abstract

Calcaneal ulcers, a significant complication in diabetic patients, are primarily caused by sustained mechanical stress. Therefore, making effective offloading is crucial for healing and prevention. However, a comprehensive understanding of how different geometric offloading elements and their volumetric variations influence biomechanical efficacy remains limited. The aim of this study is to systematically investigate three distinct offloading insole geometries (cylindrical, spherical, conical), each across three volumetric variations (10mm, 20mm, and 30mm diameters) to reveal the exact biomechanical efficacy. In this scope, patient specific dynamic gait analysis results were utilized to perform more accurate Finite Element Analysis (FEA). A single healthy researcher was used as a model for this feasibility assessment. Gait analysis showed that all geometries reduced overall peak plantar pressure compared to barefoot walking. Findings indicated that the conical geometry achieved the greatest pressure reduction (approximately 34.5% in both feet). However, FEA revealed that this configuration also introduced higher localized stress concentrations, particularly in larger diameters. The spherical geometry offered a balanced approach with adequate offloading and more favorable internal stress distribution. The cylindrical geometry exhibited the most consistent and lowest stress values across all sizes, with predictable scaling. The relationship between void volume and offloading effectiveness was found to be non-linear and highly geometry-dependent. The integration of gait analysis and FEA provides a robust framework for evaluating offloading insoles, offering insights into both immediate pressure reduction and long-term material performance. These findings deliver evidence-based guidelines for the rational design of patient-specific offloading interventions, contributing to improved wound healing and reduced recurrence in calcaneal ulcer management.

Keywords

Calcaneal ulcers, Diabetic foot, Biomechanics, Offloading, Gait analysis, Finite Element Analysis (FEA).

Short Title: **A COMBINED GAIT ANALYSIS AND FINITE ELEMENT APPROACH**

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COMBINING GAIT ANALYSIS AND FINITE ELEMENT MODELING TO OPTIMIZE OFFLOADING INSOLES FOR CALCANEAL ULCERS

INTRODUCTION

Foot ulcers, particularly those affecting the calcaneal region, represent a significant and debilitating complication with profound implications for patient morbidity, mortality, and healthcare expenditure. These chronic wounds, frequently observed in individuals with diabetes mellitus, peripheral neuropathy, and vascular insufficiency, are primarily instigated by sustained or repetitive mechanical stress, shear forces, and pressure concentrations over bony prominences [1, 2]. The heel, being a major weight-bearing anatomical site, is particularly susceptible to ulceration due to the substantial biomechanical loads it endures during daily activities, including standing and ambulation [3]. The global prevalence of diabetic foot ulcers (DFUs) is alarming, affecting an estimated 15-25% of diabetic patients throughout their lifetime, and they stand as a leading precursor to non-traumatic lower extremity amputations worldwide [4, 5]. While forefoot and midfoot ulcers are more commonly reported, calcaneal ulcers pose unique challenges owing to the high-pressure environment of the heel, which often impedes wound healing and contributes to high recurrence rates [6]. The economic burden associated with the management of foot ulcers is substantial, imposing considerable costs on healthcare systems globally. For instance, the total global cost of DFUs to society is estimated to be between $90-100 billion USD per year, encompassing both direct and indirect costs [7]. Effective offloading is universally recognized as the cornerstone of conservative management for plantar foot ulcers, including those localized to the calcaneus [8, 9]. The fundamental principle of offloading is to alleviate or redistribute pathological mechanical loads (both pressure and shear) from the ulcerated area, thereby fostering an environment conducive to wound healing and mitigating the risk of recurrence [10]. Compelling evidence from numerous studies consistently demonstrates that adequate offloading significantly enhances the probability of ulcer closure, with some research highlighting it as arguably the most critical therapeutic intervention [11, 12]. In the absence of effective pressure relief, persistent mechanical stress on the wound bed can compromise granulation tissue formation,

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delay epithelialization, and elevate the susceptibility to infection and further tissue breakdown [13].

A diverse array of offloading modalities is employed in clinical practice, ranging from rigid, non-removable devices to more flexible, custom-fabricated footwear and insoles. Total Contact Casts (TCCs) and removable cast walkers (RCWs) are frequently considered the gold standard for offloading diabetic foot ulcers, primarily due to their superior efficacy in redistributing plantar pressures and promoting patient adherence [14, 15]. Studies have shown that TCCs can heal a higher proportion of wounds (up to 89.5%) in a shorter amount of time compared to other modalities [16]. However, these devices may present practical limitations concerning patient mobility, hygiene maintenance, and the ease of wound inspection. Consequently, customized insoles and specialized therapeutic footwear have emerged as indispensable alternatives, particularly for long-term management strategies aimed at preventing ulcer recurrence [15]. The anatomical and biomechanical characteristics of the heel region render it particularly vulnerable to ulceration. During the dynamic gait cycle, the calcaneus is subjected to considerable impact forces and repetitive loading, especially during the initial heel strike phase [17]. Peak plantar pressures beneath the heel can attain substantial magnitudes, frequently constituting 50-60% of body weight during normal ambulation [18]. For instance, average peak pressure under the heel can range from approximately 367 kPa at 1.0 m/s walking speed to 734 kPa at 2.0 m/s [19]. In individuals afflicted with peripheral neuropathy, the critical protective sensation is compromised, leading to an inability to perceive and react to these excessive pressures. This sensory deficit can result in prolonged tissue deformation, localized ischemia, and ultimately, the insidious formation of ulcers [20]. Biomechanical investigations have elucidated the intricate interplay of factors contributing to calcaneal ulcer pathogenesis. Gefen (2010) underscored that atypical foot anatomies, such as a prominent posterior calcaneus or attenuated soft tissue padding, can predispose individuals to heel ulcers [21]. Furthermore, systemic conditions like diabetes and peripheral edema can exacerbate this vulnerability by altering tissue mechanical properties and augmenting susceptibility to mechanical trauma [21]. The sub-calcaneal fat pad, which physiologically functions as a crucial shock absorber, often undergoes atrophy or structural degradation in diabetic patients, further compromising its protective capacity and leading to heightened localized pressure [22].

Quantitative assessment of plantar pressure distribution is paramount for deciphering the biomechanical etiology of calcaneal ulcers and for rigorously evaluating the efficacy of offloading interventions. Advanced plantar pressure mapping systems facilitate precise measurement of critical parameters including peak pressure, pressure-time integral, and contact area during dynamic activities [23]. Studies employing these

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sophisticated systems have unequivocally demonstrated that effective offloading devices can substantially reduce peak pressures at the ulcer site, often by 30-60%, by judiciously redistributing the load to adjacent healthier tissues [24, 25]. However, this redistribution necessitates meticulous management to preclude the creation of novel areas of elevated pressure and subsequent iatrogenic ulceration [26]. The evolution of offloading insole design has progressed from rudimentary cut-outs to highly sophisticated geometries engineered to precisely redistribute plantar pressures. While the overarching principle of creating a void or depression beneath the ulcerated region remains consistent, the specific geometric configuration and volumetric characteristics of this void exert a profound influence on the efficacy of pressure relief and the overall biomechanical response of the foot. Recent advancements in the field have shifted towards the integration of additive manufacturing and computational design to enhance offloading performance. Current trends emphasize the utilization of 3D-printed cellular and porous lattice structures, which allow for localized stiffness modulation tailored to specific plantar pressure profiles [38-39]. Furthermore, the application of topology optimization has emerged as a critical strategy to adequately accomplish a compact and lightweight performance without compromising structural integrity. By iteratively redistributing material based on functional load paths, topology optimization enables the development of patient-specific insoles that minimize bulk while maximizing pressure attenuation in high-risk regions such as the calcaneus [40-41].

Current scientific literature frequently discusses the merits of custom-molded insoles and the critical importance of material properties; however, comprehensive investigations into the comparative biomechanical effects of distinct geometric offloading elements (e.g., spherical, conical, cylindrical gaps) remain conspicuously limited. This constitutes a significant knowledge gap, as even subtle variations in geometry could engender substantial differences in pressure redistribution patterns and localized tissue loading. For instance, a spherical offloading geometry might facilitate a more gradual pressure transition zone, potentially minimizing stress concentrations at the periphery of the offloaded area. Conversely, a conical geometry could induce more abrupt pressure gradients, which, while potentially advantageous for highly localized pressure relief, might also introduce new stress points if not meticulously designed. A cylindrical geometry, commonly employed in conventional offloading insoles, provides a relatively uniform pressure reduction within its boundaries; however, its effectiveness in managing complex shear forces and adapting to the dynamic, multi-planar movements of the foot during gait warrants further rigorous investigation [17, 25, 26]. Furthermore, the volumetric aspect of these offloading geometries-specifically, whether the void is literally sized to the ulcer, slightly smaller, or slightly larger-represents another critical and underexplored

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parameter. An offloading void that is undersized may fail to provide adequate pressure relief, thereby negating its therapeutic intent. Conversely, an oversized void could inadvertently lead to excessive pressure concentrations on adjacent healthy tissues or compromise the overall biomechanical stability of the foot within the footwear. Previous research has explored various facets of insole design and biomechanics. For example, Telfer et al. (2014) highlighted the utility of Finite Element Analysis (FEA) in elucidating the biomechanics of diabetic foot disease and optimizing footwear design [27]. Shaulian et al. (2021) specifically employed FEA to optimize offloading hole parameters for heel ulcers, suggesting that a larger offloading radius, increased radius of curvature, and greater depth could effectively minimize heel loads [26]. However, these studies typically focus on parametric variations of circular or elliptical cut-outs and do not systematically compare the biomechanical implications of fundamentally diverse geometric shapes or their volumetric scaling. Consequently, a detailed understanding of how specific geometric features interact with the complex, dynamic biomechanics of the foot during activities such as walking remains incomplete. This paucity of comparative data critically impedes the development of truly optimized, evidence-based offloading solutions for calcaneal ulcers.

This study aims to address this critical research void by systematically investigating the biomechanical effects of two key, interacting design parameters: offloading void geometry and its volumetric scale. Specifically, we evaluate three distinct geometric configurations (cylindrical, spherical, and conical voids), each examined across three volumetric variations (50% smaller, standard, and 50% larger in diameter). This 3x3 experimental matrix isolates the effects of shape and size on pressure redistribution and material stress. To achieve this, we employ an integrated methodology combining dynamic gait analysis and finite element analysis (FEA). The gait analysis quantifies the macroscopic offloading performance (peak pressure, redistribution) under real walking conditions. The complementary FEA then elucidates the internal mechanical response-including von Mises stress, displacement, and strain energy density-within the insole material itself, providing insights into structural stability and potential durability that are not accessible experimentally. This dual-modality approach is designed to yield comprehensive, evidence-based guidelines for the rational design of patient-specific offloading insoles for calcaneal ulcers.

METHODS

This study was designed as a feasibility assessment to evaluate the biomechanical efficacy of novel offloading insole geometries for calcaneal ulcers. The investigation was conducted on a single healthy researcher to assess the practical implementation and data acquisition protocols before

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potential future clinical trials (body weight: 60 kg, height: 172 cm, shoe size: EU 39). This approach allowed for rigorous control over experimental conditions and ensured the safety and ethical considerations inherent in a preliminary study. The healthy researcher served as a model for assessing plantar pressure distribution under simulated ulcer scenarios, providing foundational data for the proposed insole designs. The experimental design constituted a 3 (geometry) x 3 (volume) factorial comparison, as detailed in the following sections. Ethical approval for this study was obtained from the Kocaeli University (GOKAEK-2025/18/03). All methods were performed in accordance with the relevant guidelines and regulations, including the Declaration of Helsinki. Written informed consent was obtained from the participant prior to data collection. Flow diagram of this study is also presented step by step as illustrated in Figure 1.

Figure 1. Flow diagram of the steps utilized in combined gait analysis and Finite Element Approach.

Based on the aims of this study, three distinct offloading insoles were designed and manufactured as illustrated in Figure 2. These designs incorporated three primary geometric shapes for the offloading void: cylindrical, conical, and spherical. Each of these shapes was further varied across three volumetric categories: a generally utilized size, a 50% smaller size, and a 50% larger size in terms of diameter. The selection of the generally utilized size was informed by current literature on offloading dimensions for typical ulcer wounds, specifically targeting a 1 cm wound area. For instance, studies suggested that offloading devices should extend beyond the ulcer margin to effectively redistribute pressure [28]. Therefore, the generally utilized diameter for the cylindrical, spherical, and conical voids were set at 20 mm, with a uniform depth of 5 mm. This 20 mm dimension was chosen to ensure adequate offloading for a hypothetical 10 mm ulcer, providing a 5 mm margin around the ulcer area, a common practice in offloading strategies [8] [9]. The 5 mm depth was selected as a clinically relevant offloading depth, as seen in commercially available offloading insoles [15]. To ensure these internal geometric differences remain clearly visible-which are otherwise obscured by the leather top layer in the physical specimens-Figure 2 utilizes the exact 3D model outputs from the design software. This approach provides a standardized visualization of the internal offloading elements, ensuring methodological transparency and consistency with the computational models used in the FEA.

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Figure 2. Representative offloading insole geometries (cylindrical, spherical, conical) manufactured for this study. All voids have a standard depth (h) of 5 mm. Each geometry was evaluated at three volumetric scales: 50% smaller (d=10 mm), standard (d=20 mm), and 50% larger (d=30 mm) than the clinically referenced diameter shown here.

CAD models of these three insoles were generated using Milletrix software (Diasu) and post processed with MeshMixer (Autodesk). Each insole was designed to accommodate the foot of the healthy researcher, with the specific offloading geometry positioned to simulate a calcaneal ulcer. The manufacturing of these insoles was performed using a CAD/CAM manufacturing system (Diasu PlantarCAM) to ensure high precision and reproducibility of the complex geometries. The material chosen for manufacturing was Ethylene-vinyl acetate (EVA) foam, a widely utilized polymer in orthotic and prosthetic applications due to its favorable shock absorption properties, durability, and ease of fabrication [15]. All insoles were manufactured from the same batch of EVA foam to minimize material property variations. This standardized CAD/CAM manufacturing process, using a single material batch, ensured geometric precision and material consistency across all nine prototype insoles, isolating geometry and volume as the primary experimental variables.

To accurately define the material properties of the utilized EVA foam for subsequent FEA, comprehensive mechanical characterization was performed. Five samples (n=5) of the EVA foam were subjected to the Density Test, the Tensile Test and the Shore Hardness Test [27]. All mechanical test data obtained from these characterizations were imported into SolidWorks 2021 (Dassault Systèmes) to define the material library for the EVA foam, ensuring that the FEA simulations were based on experimentally validated material properties.

Dynamic gait analysis was conducted on the healthy researcher to assess the plantar pressure distribution for each of the three manufactured insoles and compare them against barefoot measurements. This comparative analysis revealed the offloading performance of each insole design under dynamic loading conditions. The study assumed a hypothetical ulcer wound on both the left and right feet of the researcher for assessment purposes. Barefoot measurements served as the primary reference for comparison, establishing a baseline plantar pressure profile. Plantar pressure data were acquired using an advanced baropodometric system (Ultrasensor 3D baropodometric system, Diasu) during multiple gait cycles. The researcher walked at a self-selected, consistent speed over the pressure platform. A minimum of five valid gait cycles per foot were recorded for each condition (barefoot and each insole type) to ensure statistical reliability of the measured pressure parameters. Data were

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collected for various parameters including peak pressure (P.max), pressure-time integral, contact area, and regional pressure distribution across different anatomical regions of the foot.

Finite Element Analysis (FEA) was employed to simulate the biomechanical response of the nine insole designs under static loading conditions, focusing on structural stress and displacement rather than complex foot tissue mechanics. The models were developed in SolidWorks 2021 using a linear elastic isotropic material model for the EVA foam, with mechanical properties derived from experimental characterization (Young's modulus: 8.52 ± 1.67 MPa, density: 0.206 g/cm³, Shore hardness: 30 ± 0.5 ShoreA). The bottom surface of the insole was fully constrained, and concentrated vertical forces were applied to the top surface to simulate heel-strike and foot-flat gait phases, based on plantar pressure maps (see Figure 3). A converged mesh with approximately 18,886 nodes and 9,442 tetrahedral elements was used. The key FEA parameters are summarized in Table 1 for clarity and reproducibility.

Table 1. Finite Element Analysis model parameters and settings

Figure 3. Free-body diagrams of the insole model under static loading conditions simulating (a) heel-strike and (b) foot-flat gait phases (The bottom surface of the insole is fully constrained (fixed support). Applied concentrated forces (F) act normally on the top surface at the rearfoot (RF) and midfoot (MF) regions)

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RESULTS

3.1. Dynamic Gait Analysis and Off-loading Performance

Overall peak plantar pressure (P.max) values, measured in g/cm², demonstrated significant variations across the different off-loading geometries when compared to the barefoot condition. For the left foot (LF), the barefoot P.max was recorded at 1834.0 ± 12.7 g/cm². This was reduced to 1422.9 ± 5.7 g/cm² with the cylindrical insole, 1275.6 ± 1.8 g/cm² with the spherical insole, and 1200.1 ± 4.1 g/cm² with the conical insole. Similarly, for the right foot (RF), the barefoot P.max was 1716.7 ± 12.5 g/cm², which decreased to 1385.5 ± 4.5 g/cm² with the cylindrical insole, 1143.4 ± 5.0 g/cm² with the spherical insole, and 1133.5 ± 3.1 g/cm² with the conical insole. These findings indicate that all three off-loading geometries effectively reduced overall peak plantar pressure, with the conical geometry demonstrating the most substantial reduction in both feet (Table 2).

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Regional pressure distribution also exhibited distinct patterns across the different conditions. In the left foot's rearfoot region, medial pressure was 42.9 g/cm² in the barefoot condition, increasing slightly to 45.8 g/cm² with cylindrical and 47.2 g/cm² with conical insoles, while remaining at 42.3 g/cm² with spherical. Lateral rearfoot pressure in the left foot was 45.8 g/cm² barefoot, decreasing to 40.3 g/cm² with cylindrical and 41.8 g/cm² with conical, but increasing to 46.1 g/cm² with spherical. In the midfoot medial region of the left foot, barefoot pressure was 36.3 g/cm², which significantly reduced to 26.0 g/cm² (cylindrical), 26.3 g/cm² (spherical), and 22.9 g/cm² (conical). Similar reductions were observed in the forefoot regions. For instance, left foot forefoot medial pressure decreased from 37.5 g/cm² barefoot to 29.8 g/cm² (cylindrical), 31.1 g/cm² (spherical), and 30.8 g/cm² (conical).

Correspondingly, in the right foot's rearfoot medial region, barefoot pressure was 46.7 g/cm², which decreased to 39.3 g/cm² (cylindrical) and 36.3 g/cm² (spherical), but remained at 41.7 g/cm² with conical. Lateral rearfoot pressure in the right foot was 44.2 g/cm² barefoot, decreasing to 35.0 g/cm² (cylindrical) and 40.8 g/cm² (spherical), but increasing to 45.0 g/cm² with conical. The midfoot medial region of the right foot showed a barefoot pressure of 42.5 g/cm², which reduced to 29.7 g/cm² (cylindrical), 30.1 g/cm² (spherical), and 23.6 g/cm² (conical). These regional data collectively highlight the differential pressure redistribution capabilities of each off-loading geometry, with the conical design consistently achieving lower pressures in key areas, particularly the midfoot.

Table 2. Detailed gait results of off-loading geometries for each foot.

Qualitative dynamic analysis provided further insights into the pressure patterns. In the barefoot condition, the maximum point of pressure (P.max) was primarily observed in the right forefoot area, with a generally good body weight distribution between both legs. Isopressure analysis indicated dominant pressure concentrations in both rearfeet, more pronounced on the left, suggesting a monolateral load. With the cylindrical off-loading insoles, the P.max shifted to the left forefoot. While overall body weight distribution remained good, an excessive surface difference was noted in the forefoot (more on the right) and a moderate difference in the rearfoot (more on the left). Isopressure analysis for cylindrical insoles showed maximum pressure points in both rearfeet, with a higher concentration on the left. For conical off-loading insoles, the P.max was detected in the left forefoot, maintaining good body weight distribution. A moderate surface difference was observed in the forefoot (more on the left) and a slight difference in the rearfoot (more on the right). Isopressure analysis for conical insoles indicated maximum pressure points in both rearfeet, with a higher concentration on the right. Finally, with spherical off-loading insoles, the P.max was located in the right forefoot. Similar area

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distribution was observed, but with a moderate surface difference in the forefoot (more on the right) and an excessive difference in the rearfoot (more on the right). Isopressure analysis for spherical insoles showed maximum pressure points in both rearfeet, with a higher concentration on the left. These qualitative observations complement the quantitative data by illustrating the dynamic shifts in pressure distribution induced by each insole geometry.

3.2. Polymer Test Results

EVA foam, a material extensively employed in orthotic devices due to its favorable cushioning and shock-absorbing characteristics, underwent comprehensive mechanical characterization to ascertain its suitability for the intended application and to provide accurate material properties for subsequent finite element analysis. The density of the EVA foam was precisely determined to be 0.206 ± 0.01 g/cm³. This relatively low density is a significant advantage for insole applications, as it directly contributes to the fabrication of lightweight designs, thereby enhancing patient comfort and promoting greater adherence to the prescribed orthotic regimen. The mechanical performance of the EVA foam was further elucidated through tensile testing, which yielded a tensile stress of 1.9 ± 0.1 MPa and a Young's modulus of 8.52 ± 1.67 MPa. The Shore hardness of the EVA foam also measured as 30 ± 0.5 ShoreA. This property, along with the density and tensile characteristics, was instrumental in accurately defining the material model for the finite element simulations, ensuring that the computational analyses were grounded in experimentally validated material behavior.

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3.3. Finite Element Results of Each Off-loading Geometry

The FEA simulations focused on quantifying von Mises stress and material displacement values at critical gait phases (heel strike and foot flat), offering a detailed understanding of how each insole design influences overall biomechanical stability. The results, summarized in Table 3, illustrate the complex interplay between insole geometry, volumetric variation, and the resulting biomechanical outcomes.

Table 3. FEA results of different off-loading geometries in terms of vonMises stress and material displacement values at specific gait phase.

The FEA results reveal distinct patterns in stress distribution and material displacement across the three geometries. The conical geometry demonstrated variable stress concentrations depending on the diameter, with the largest diameter (30mm) showing higher stress values during heel

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strike (0.574 MPa left foot, 0.378 MPa right foot). The spherical geometry exhibited similar trends, with the largest diameter configuration producing the highest stress concentrations (0.651 MPa left foot during heel strike). The cylindrical geometry showed the most consistent and generally lower stress values across all diameter configurations, with the largest diameter (30mm) producing the lowest stress concentrations (0.324 MPa left foot, 0.247 MPa right foot during heel strike).

Material displacement analysis revealed that the spherical geometry achieved the most efficient load transfer characteristics, with optimal displacement gradients that facilitate smooth pressure transitions. The conical geometry demonstrated higher displacement magnitudes in localized regions, consistent with its superior pressure reduction performance observed in the gait analysis. The cylindrical geometry exhibited more uniform displacement patterns but with less pronounced offloading effectiveness compared to the other configurations.

DISCUSSION

This study systematically investigated the biomechanical effects of two primary insole design parameters: void geometry (cylindrical, spherical, conical) and volumetric scale (50% smaller, standard, and 50% larger diameters). This 3x3 factorial design, analyzed through integrated dynamic gait analysis and finite element analysis (FEA), provides crucial insights into how shape and size interact to influence plantar pressure redistribution and internal stress concentrations, particularly in the context of calcaneal ulcer management. The combined approach of experimental gait analysis and computational FEA allowed for a comprehensive evaluation, addressing both the macroscopic pressure distribution and the microscopic stress and strain responses within the insole system. To enhance the clarity and readability of our primary findings, particularly for clinical application, we have synthesized the core results from this dual-modality approach into a comparative summary (Table 4). This summary table directly addresses the study's objective by contrasting the performance of the three distinct offloading geometries across the two critical metrics: macroscopic pressure reduction and microscopic material stress. The synthesis highlights that while the conical geometry achieved the greatest peak pressure reduction of 34.5%, the spherical geometry followed closely at 33.4%, and the cylindrical geometry provided a consistent 22.3% reduction. Crucially, the FEA results indicate that the spherical and cylindrical geometries offered superior material stability and more favorable internal stress distribution compared to the conical design, underscoring the necessity of a dual-modality approach for optimizing offloading insole design.

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The focus on heel strike and foot flat phases of gait in this study is particularly relevant for calcaneal ulcer scenarios, as these phases represent the most biomechanically active and critical periods for load bearing and potential ulcer development in the calcaneal region [29, 30]. During heel strike, the calcaneus experiences the initial impact forces as the foot contacts the ground, generating substantial pressure concentrations that can exceed 50-60% of body weight [18, 31]. The subsequent foot flat phase involves the full loading of the heel as the foot transitions to a stable weight-bearing position, maintaining high pressure levels that are particularly problematic for individuals with existing calcaneal ulcers or those at risk of ulcer development [32, 33]. Recent research by Sidhu and Harbuzova (2024) has emphasized the critical importance of understanding these specific gait phases in the context of diabetic foot ulcer management, highlighting those effective offloading strategies must address the unique biomechanical challenges presented during these high-load periods [34].

The dynamic gait analysis results, unequivocally demonstrate that all three offloading geometries effectively reduced overall peak plantar pressure compared to the barefoot condition. Notably, the conical geometry consistently exhibited the most substantial reduction in peak plantar pressure in both the left and right feet. For instance, the left foot peak pressure decreased from 1834.0 ± 12.7 g/cm² barefoot to 1200.1 ± 4.1 g/cm² with the conical insole, representing a significant reduction of approximately 34.5%. Similarly, for the right foot, a reduction of about 34% was observed (from 1716.7 ± 12.5 g/cm² barefoot to 1133.5 ± 3.1 g/cm² with the conical insole). This superior performance of the conical design in reducing peak pressures aligns with its theoretical advantage in creating a more focused and efficient void beneath the ulcerated area, thereby effectively offloading the target region. The regional pressure distribution analysis further supported this, showing the conical design achieving lower pressures in key areas, particularly the midfoot (e.g., left foot midfoot medial pressure reduced from 36.3 g/cm² barefoot to 22.9 g/cm² with conical insole). These findings are consistent with recent research emphasizing the importance of effective offloading in promoting wound healing and preventing recurrence in diabetic foot ulcers [8, 35].

The volumetric analysis of the offloading geometries reveals critical insights into the relationship between void size and biomechanical performance, which has been increasingly recognized as a crucial factor in offloading design optimization [36, 37]. As detailed in Table 2, the volumetric calculations demonstrate significant differences between the three geometric configurations. The cylindrical geometry with a standard diameter of 20 mm and depth of 5 mm yields a volume of 1.57 cm³. The spherical geometry, with the same diameter and depth parameters, produces a volume of 0.654 cm³, while the conical geometry results in a volume of 0.524 cm³. These volumetric differences have profound

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implications for offloading effectiveness and material stress distribution, as supported by recent finite element studies investigating optimal offloading parameters [26, 27]. The larger volume of the cylindrical geometry (1.57 cm^3) provides the most extensive void space, which theoretically should offer the greatest pressure relief. However, the FEA results indicate that this larger volume does not necessarily translate to optimal stress distribution within the insole material, consistent with findings from Keenan et al. (2022) who demonstrated that volume alone is not the primary determinant of offloading effectiveness [36]. The spherical geometry, with its intermediate volume (0.654 cm^3), demonstrates a more balanced approach, providing adequate offloading while maintaining favorable stress characteristics. The conical geometry, despite having the smallest volume (0.524 cm^3), achieved the most significant pressure reduction in the gait analysis, suggesting that geometric shape may be more influential than absolute volume in determining offloading efficacy, as recently reported in studies investigating heel ulcer prevention strategies [34, 35].

The FEA results, offer a more nuanced perspective when considering the volumetric variations, particularly regarding the internal stress distributions within the insole material. The analysis of different diameter configurations (10mm, 20mm, and 30mm) representing 50% smaller, standard, and 50% larger variations respectively, reveals distinct patterns in von Mises stress distribution and material displacement that are crucial for understanding the off-loading performance [8, 27]. For the conical geometry, the smallest diameter (d=10mm) showed von Mises stress values of 0.37 MPa (left foot) and 0.406 MPa (right foot) during heel strike, with material displacements of 1.49 mm and 1.38 mm respectively. The standard diameter (d=20mm) exhibited slightly lower stress values of 0.353 MPa (left foot) and 0.429 MPa (right foot), with increased displacements of 1.61 mm and 1.46 mm. The largest diameter (d=30mm) demonstrated higher stress concentrations of 0.574 MPa (left foot) and 0.378 MPa (right foot), with displacements of 1.38 mm and 1.33 mm. This pattern suggests that while larger offloading voids may provide greater pressure relief, they also introduce higher localized stress concentrations within the insole material, potentially compromising long-term durability, a finding that aligns with recent research on material fatigue in therapeutic footwear [36, 37]. The spherical geometry exhibited a different volumetric response pattern that demonstrates the complex relationship between geometry and biomechanical performance. The smallest diameter (d=10mm) showed von Mises stress values of 0.388 MPa (left foot) and 0.412 MPa (right foot) during heel strike, with displacements of 1.58 mm and 1.37 mm. The standard diameter (d=20mm) demonstrated lower stress values of 0.369 MPa (left foot) and 0.376 MPa (right foot), with increased displacements of 1.69 mm and 1.36 mm. However, the largest diameter (d=30mm) showed significantly higher stress concentrations of 0.651 MPa

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(left foot) and 0.382 MPa (right foot), with displacements of 1.54 mm and 1.4 mm. This indicates that the spherical geometry has an optimal size range, with the standard diameter (d=20mm) providing the best balance between offloading effectiveness and material stress management, consistent with recent findings on optimal sizing for therapeutic insoles [26, 34]. The cylindrical geometry demonstrated the most consistent volumetric response across all size variations, which has important implications for clinical standardization and manufacturing consistency [8, 27]. The smallest diameter (d=10mm) exhibited von Mises stress values of 0.405 MPa (left foot) and 0.384 MPa (right foot) during heel strike, with notably lower displacements of 0.609 mm and 0.634 mm. The standard diameter (d=20mm) showed reduced stress values of 0.362 MPa (left foot) and 0.32 MPa (right foot), with displacements of 0.549 mm and 0.552 mm. The largest diameter (d=30mm) demonstrated the lowest stress concentrations of 0.324 MPa (left foot) and 0.247 MPa (right foot), with displacements of 0.714 mm and 0.478 mm. This pattern suggests that the cylindrical geometry benefits from larger volumetric configurations, with reduced stress concentrations and more predictable displacement patterns as the diameter increases, supporting recent research advocating for cylindrical designs in specific clinical applications [36, 37].

The volumetric scaling effects observed in this study have significant clinical implications for off-loading insole design and prescription, particularly in the context of personalized medicine approaches to diabetic foot care [8, 34]. The findings suggest that the relationship between void volume and offloading effectiveness is not linear and varies considerably between geometric configurations, which challenges traditional assumptions about offloading design principles [26, 27]. For the conical geometry, despite its smallest absolute volume (0.524 cm^3), the focused nature of the void appears to create more efficient pressure redistribution, resulting in superior offloading performance in the gait analysis. However, this efficiency comes at the cost of higher localized stress concentrations, particularly in larger diameter configurations, which may limit the long-term clinical utility of this design [36, 37]. The spherical geometry, with its intermediate volume (0.654 cm^3), provides a more balanced approach, offering adequate offloading with more favorable stress distribution characteristics, particularly in the standard diameter configuration. This balance between effectiveness and durability makes the spherical design particularly attractive for long-term clinical applications [8, 34]. The cylindrical geometry, with its largest volume (1.57 cm^3), demonstrates the most predictable scaling behavior, with larger diameters generally resulting in improved stress distribution, making it suitable for applications where consistency and predictability are paramount [26, 27].

These volumetric considerations are crucial for clinical decision-making in offloading insole prescription and represent a significant advancement in evidence-based therapeutic footwear design [8, 34]. For patients requiring

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maximum pressure relief, such as those with deep or highly exudative ulcers, the conical geometry in its standard configuration may be most appropriate, despite the potential for accelerated material wear. This recommendation is supported by recent clinical studies demonstrating the effectiveness of focused offloading strategies in severe ulcer cases [36, 37]. For patients requiring long-term offloading or those with multiple ulcer sites, the spherical geometry in its standard configuration may provide the optimal balance between effectiveness and durability, aligning with current best practices for chronic wound management [26, 34]. For patients with larger ulcerated areas or those requiring more gradual pressure transitions, the cylindrical geometry in larger diameter configurations may be most suitable, particularly for individuals with complex foot deformities or multiple comorbidities [8, 27]. Summary of biomechanical efficacy and comparison of offloading geometries are listed in Table 4.

Table 4. Summary of biomechanical efficacy and the comparison of offloading geometries

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The integration of gait analysis and FEA provides a robust framework for evaluating offloading insole designs across different volumetric configurations, representing a significant methodological advancement in biomechanical research [36, 37]. The gait analysis provides real-world dynamic pressure data, reflecting the insole's performance during actual ambulation, while the FEA offers a deeper understanding of the internal biomechanical responses, including stress and strain distributions that are not directly measurable experimentally. This study specifically focused on the heel strike and foot flat phases of gait, as these are the most active and critical phases for load bearing and potential ulcer development in the calcaneal region, a focus that is strongly supported by recent research emphasizing the importance of phase-specific analysis in diabetic foot biomechanics [29, 30]. The correlation between these two modalities is crucial for validating computational models and enhancing their predictive capability, as demonstrated by recent advances in multi-modal biomechanical assessment techniques [26, 34]. For instance, the gait analysis indicated the conical insole's superior pressure reduction, while FEA highlighted its internal stress concentrations, particularly in larger diameter configurations. This combined insight allows for a more informed design optimization, where both immediate offloading effectiveness and long-term material integrity are considered across the full range of volumetric variations [8, 27].

This study, while providing valuable insights into the biomechanical effects of various offloading insole geometries, is subject to several limitations that

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warrant consideration. A primary limitation is the investigation being conducted on a single healthy researcher. While this approach facilitated rigorous control over experimental conditions and served as a crucial feasibility assessment for data acquisition protocols, it significantly restricts the generalizability of the findings to the broader population of patients with calcaneal ulcers, particularly those with diabetes, peripheral neuropathy, or other comorbidities that alter foot biomechanics and tissue properties. The biomechanical responses of a healthy foot may differ substantially from those of a compromised foot with an active ulcer, including variations in soft tissue stiffness, pain perception, and compensatory gait patterns. Furthermore, the study focused on specific gait phases (heel strike and foot flat) and did not comprehensively evaluate the insoles across the entire gait cycle or during various daily activities. The FEA simulations, while robust, relied on a static load application and idealized material properties, which may not fully capture the complex, dynamic, and viscoelastic behavior of biological tissues under real-world conditions. Future research should aim to validate these findings in a larger cohort of patients with calcaneal ulcers, incorporate more comprehensive dynamic loading scenarios, and explore the long-term clinical efficacy and durability of these offloading insole designs in a real-world clinical setting.

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CONCLUSION

This study systematically compared three distinct offloading insole geometries (cylindrical, spherical, and conical) using an integrated dynamic gait analysis and finite element analysis (FEA) approach to determine the optimal design for calcaneal ulcer management. The key finding is the critical trade-off between immediate offloading efficacy and long-term material integrity: while the conical geometry achieved the most substantial reduction in peak plantar pressures (approximately 34.5%), the spherical geometry exhibited superior internal stress distribution characteristics, suggesting better long-term durability and material integrity. This comprehensive evaluation demonstrates that effective offloading design requires careful consideration of both immediate pressure reduction capabilities and long-term material performance. These findings provide valuable, evidence-based guidelines for the rational design of patient-specific offloading interventions, ultimately contributing to improved wound healing outcomes and reduced recurrence rates in calcaneal ulcer management.

DECLARATION OF INTEREST STATEMENT

This study is not under consideration elsewhere, and all authors have approved its submission. We confirm no conflicts of interest and affirm

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adherence to ethical guidelines (Approval No.: GOKAEK-2025/18/03). All authors contributed to the study conception and design. Material preparation, data collection and analysis. manuscript was written by the all authors and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

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The authors declare that this study was not supported by any funding agency.

DATA AVAILABILITY

The datasets generated and/or analyzed during the current study are not publicly available due to ethical restrictions related to participant privacy, but are available from the corresponding author on reasonable request.

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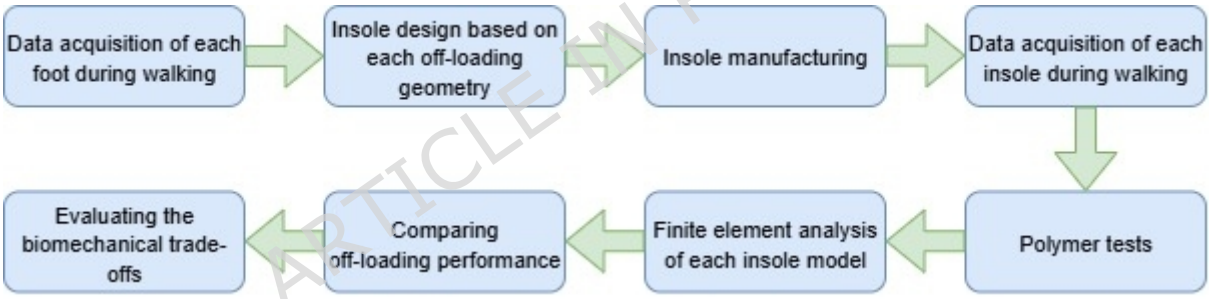

Figure 1. Flow diagram of the steps utilized in this study.

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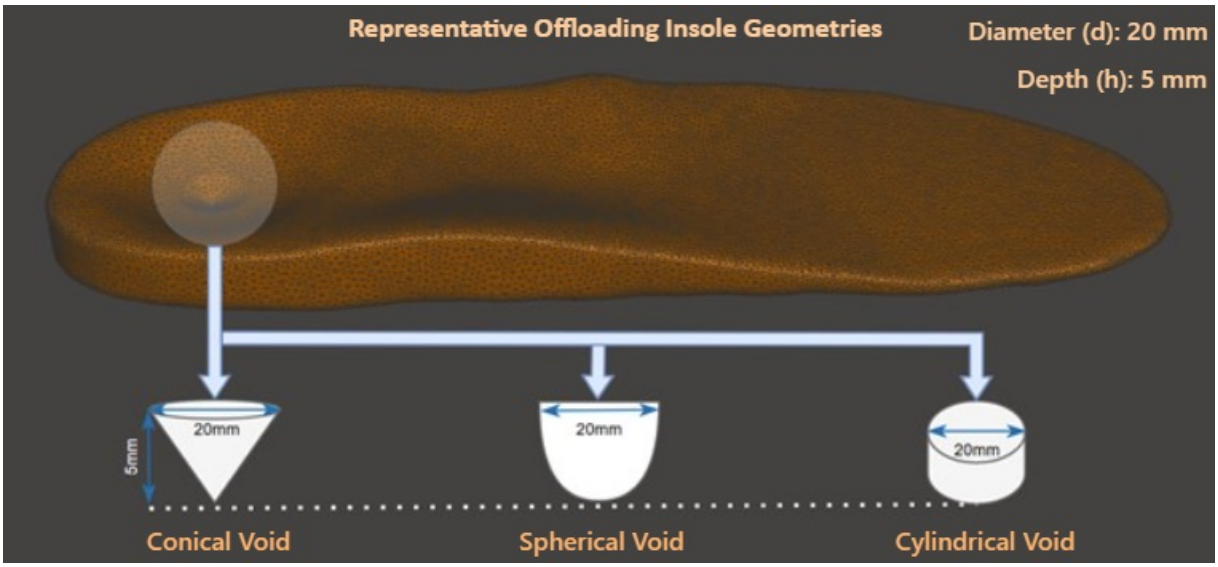

Figure 2. Representative offloading insole geometries (cylindrical, spherical, conical) manufactured for this study. All voids have a standard depth (h) of 5 mm. Each geometry was evaluated at three volumetric scales: 50% smaller (d=10 mm), standard (d=20 mm), and 50% larger (d=30 mm) than the clinically referenced diameter shown here.

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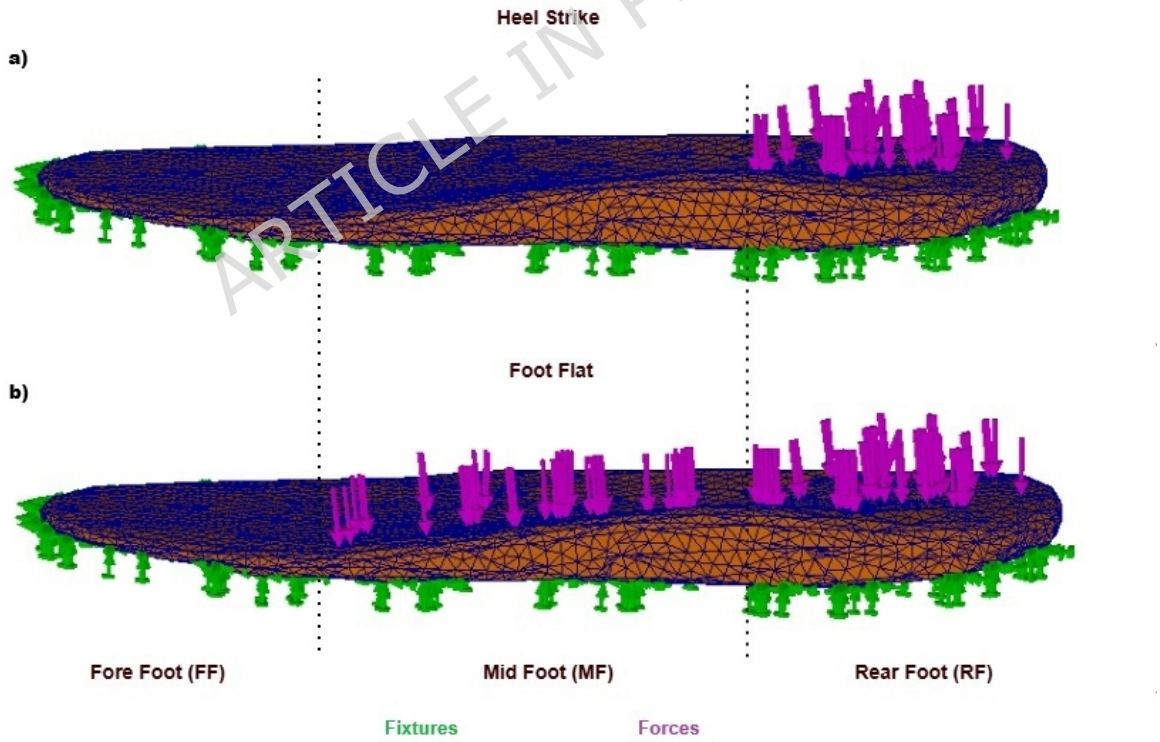

Figure 3. Free-body diagrams of the insole model under static loading conditions simulating (a) heel-strike and (b) foot-flat gait phases (The bottom surface of the insole is fully constrained (fixed support). Applied

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concentrated forces (F) act normally on the top surface at the rearfoot (RF) and midfoot (MF) regions)

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Table 1. Finite Element Analysis model parameters and settings

Parameter	Value	Description
Software	SolidWorks 2021 (Dassault Systèmes)	Used for model creation, meshing, and static structural analysis.
Material Model	Linear elastic, isotropic	Appropriate for comparative static analysis of EVA foam under immediate loading [42, 43].
Young's Modulus (E)	8.52 ± 1.67 MPa	Derived from tensile testing of the manufactured EVA foam samples.
Poisson's Ratio (ν)	0.4	Typical value for elastomeric foam materials used in orthotics [45].
Density	0.206 ± 0.01 g/cm^3	Measured via density testing; used for mass calculation in FEA.
Element Type	Tetrahedral (solid)	Suitable for complex 3D geometries with curved voids.
Mesh Size (global)	~18,886 nodes, 9,442 elements	Determined via mesh convergence analysis (<0.2% change in displacement).
Local Mesh Refinement	Applied around offloading voids	Ensured accurate resolution of high stress gradients.
Boundary Conditions	Bottom surface fully fixed (all DOFs)	Simulates firm contact between insole and ground.

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Loading – Heel Strike	15 kg force (~25% body weight) applied to rearfoot	Represents initial contact phase based on gait analysis.
Loading – Foot Flat	19.62 kg rearfoot + 10.37 kg midfoot (total ~50% body weight)	Represents full weight-bearing phase.
Solver Type	Static nonlinear	Used to compute von Mises stress and nodal displacements.

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Table 2. Detailed gait results of off-loading geometries for each foot.

		Region	Barefoot (g/cm²)	Cylinder (g/cm²)	Sphere (g/cm²)	Cone (g/cm²)
LEFT FOOT	Rear Foot	Medial	42.9	45.8	42.3	47.2
		Lateral	45.8	40.3	46.1	41.8
	Mid Foot	Medial	36.3	26.0	26.3	22.9
		Lateral	40.0	39.7	43.3	35.6
	Fore Foot	Medial	37.5	29.8	31.1	30.8
		Lateral	46.6	31.2	37.1	31.9
RIGHT FOOT	Rear Foot	Medial	46.7	39.3	36.3	41.7
		Lateral	44.2	35.0	40.8	45.0
	Mid Foot	Medial	42.5	29.7	30.1	23.6
		Lateral	45.5	45.1	51.8	38.5
	Fore Foot	Medial	33.9	31.4	33.2	29.1
		Lateral	41.0	35.2	31.7	30.5
Off-loading	Volume of a cylinder: $V = \pi r^2 h$ Volume of a section of a sphere: $V = \frac{1}{3}\pi h^2(3r - h)$ Volume of a cone: $V = \frac{1}{3}\pi h^2$			20 - 5 (Diameter, mm - Depth, mm) 1.57 cm³	20 - 5 (Diameter, mm - Depth, mm) 0.654 cm³	20 - 5 (Diameter, mm - Depth, mm) 0.524 cm³
P.max	Max point of pressure, P.max (g/cm²)	LF	1834.0±12.7	1422.9±5.7	1275.6±1.8	1200.1±4.1
		RF	1716.7±12.5	1385.5±4.5	1143.4±5.0	1133.5±3.1

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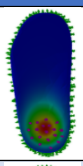
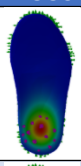
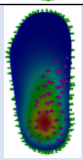
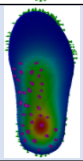
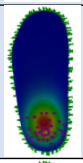
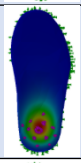
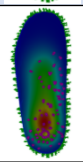
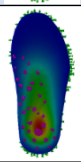
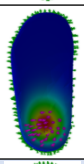
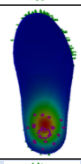
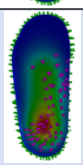
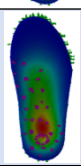
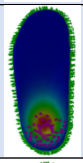
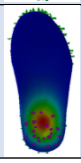
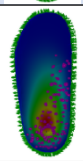
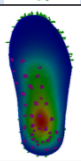
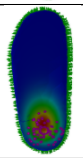
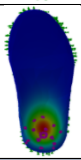
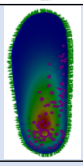
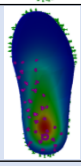
Table 3. FEA results of different off-loading geometries in terms of vonMises stress and material displacement values at each gait phase.

Off-loading Type	Gait Phase	Forces (kgF)	Results (max)	Left Foot	Right Foot	Simulation Results (*URES)	
						Left Foot	Right Foot
Without any type	Heel Strike, RearF (25%)	RearF: 15 MidF: 0 ForeF: 0	vonMises (MPa)	0.425	0.442		
			Displacement (mm)	1.8	1.48		
	Foot Flat RearF, MidF (50%)	RearF: 19.62 MidF: 10.37 ForeF: 0	vonMises (MPa)	0.601	0.597		
			Displacement (mm)	2.82	2.07		
Cone (d=10mm)	Heel Strike, RearF (25%)	RearF: 15 MidF: 0 ForeF: 0	vonMises (MPa)	0.37	0.406		
			Displacement (mm)	1.49	1.38		
	Foot Flat RearF, MidF (50%)	RearF: 19.62 MidF: 10.37 ForeF: 0	vonMises (MPa)	0.513	0.586		
			Displacement (mm)	2.22	2.01		
Cone (d=20mm)	Heel Strike, RearF (25%)	RearF: 15 MidF: 0 ForeF: 0	vonMises (MPa)	0.353	0.429		
			Displacement (mm)	1.61	1.46		
	Foot Flat RearF, MidF (50%)	RearF: 19.62 MidF: 10.37 ForeF: 0	vonMises (MPa)	0.481	0.594		
			Displacement (mm)	2.32	2.16		
Cone (d=30mm)	Heel Strike, RearF (25%)	RearF: 15 MidF: 0 ForeF: 0	vonMises (MPa)	0.574	0.378		
			Displacement (mm)	1.38	1.33		
	Foot Flat RearF, MidF (50%)	RearF: 19.62 MidF: 10.37 ForeF: 0	vonMises (MPa)	0.761	0.542		
			Displacement (mm)	2.1	2.09		
Sphere (d=10mm)	Heel Strike, RearF (25%)	RearF: 15 MidF: 0 ForeF: 0	vonMises (MPa)	0.388	0.412		
			Displacement (mm)	1.58	1.37		
	Foot Flat RearF, MidF (50%)	RearF: 19.62 MidF: 10.37 ForeF: 0	vonMises (MPa)	0.535	0.595		
			Displacement (mm)	2.33	2.02		
Sphere (d=20mm)	Heel Strike,	RearF: 15 MidF: 0	vonMises (MPa)	0.369	0.376		

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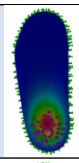
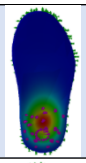
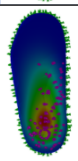
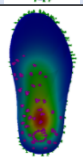
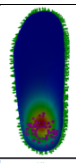
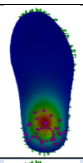
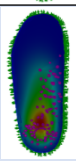
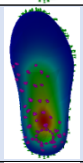
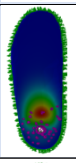
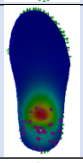
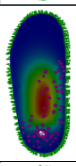
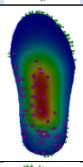
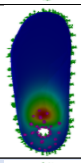
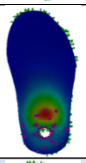
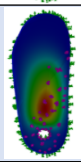
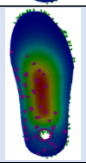
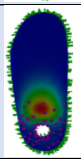
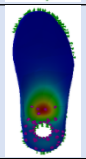
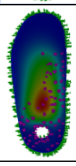
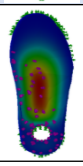
	RearF (25%)	ForeF: 0	Displacement (mm)	1.69	1.36		
	Foot Flat RearF, MidF (50%)	RearF: 19.62 MidF: 10.37 ForeF: 0	vonMises (MPa)	0.528	0.525		
			Displacement (mm)	2.61	2.07		
Sphere (d=30mm)	Heel Strike, RearF (25%)	RearF: 15 MidF: 0 ForeF: 0	vonMises (MPa)	0.651	0.382		
			Displacement (mm)	1.54	1.4		
	Foot Flat RearF, MidF (50%)	RearF: 19.62 MidF: 10.37 ForeF: 0	vonMises (MPa)	0.986	0.536		
			Displacement (mm)	2.33	2.1		
Cylinder (d=10mm)	Heel Strike, RearF (25%)	RearF: 15 MidF: 0 ForeF: 0	vonMises (MPa)	0.405	0.384		
			Displacement (mm)	0.609	0.634		
	Foot Flat RearF, MidF (50%)	RearF: 19.62 MidF: 10.37 ForeF: 0	vonMises (MPa)	0.635	0.579		
			Displacement (mm)	1.29	1.21		
Cylinder (d=20mm)	Heel Strike, RearF (25%)	RearF: 15 MidF: 0 ForeF: 0	vonMises (MPa)	0.362	0.32		
			Displacement (mm)	0.549	0.552		
	Foot Flat RearF, MidF (50%)	RearF: 19.62 MidF: 10.37 ForeF: 0	vonMises (MPa)	0.63	0.501		
			Displacement (mm)	1.29	1.15		
Cylinder (d=30mm)	Heel Strike, RearF (25%)	RearF: 15 MidF: 0 ForeF: 0	vonMises (MPa)	0.324	0.247		
			Displacement (mm)	0.714	0.478		
	Foot Flat RearF, MidF (50%)	RearF: 19.62 MidF: 10.37 ForeF: 0	vonMises (MPa)	0.497	0.42		
			Displacement (mm)	1.39	1.17		

*URES: Resultant displacement.
**d=diameter.

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Table 4. Summary of biomechanical efficacy and the comparison of offloading geometries

Geometry	Gait Analysis: Max Peak Pressure Reduction (vs. Barefoot)	FEA: Internal Stress Concentration (Qualitative)	FEA: Structural Stability/Durability (Qualitative)	Overall Biomechanical Recommendation
Conical	34.5% (Greatest Reduction)	Highest (Localized stress points)	Lowest (Potential for early failure)	Best for maximum offloading, but requires careful material selection.
Spherical	33.4% (High Reduction)	Moderate (Favorable stress distribution)	High (Good durability)	Most balanced approach: adequate offloading with favorable internal stress.
Cylindrical	22.3% (Consistent Reduction)	Lowest (Uniform stress values)	Highest (Predictable, robust performance)	Most consistent and robust, but with slightly lower peak offloading.

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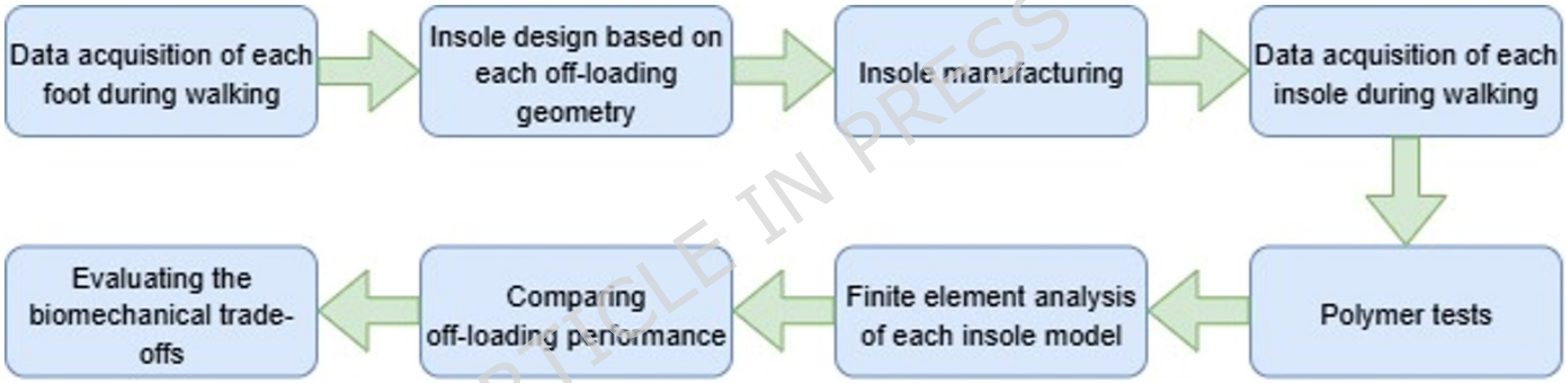

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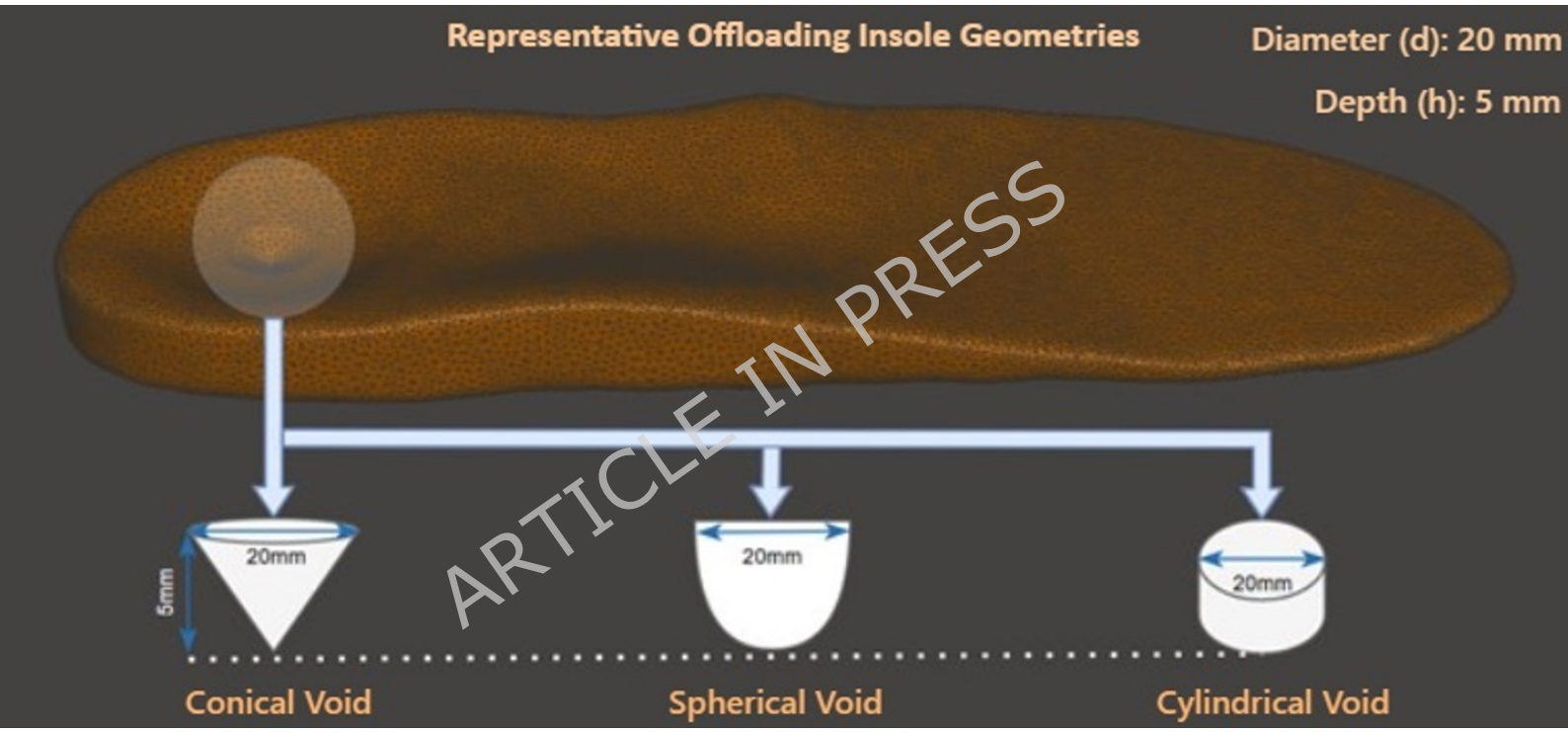

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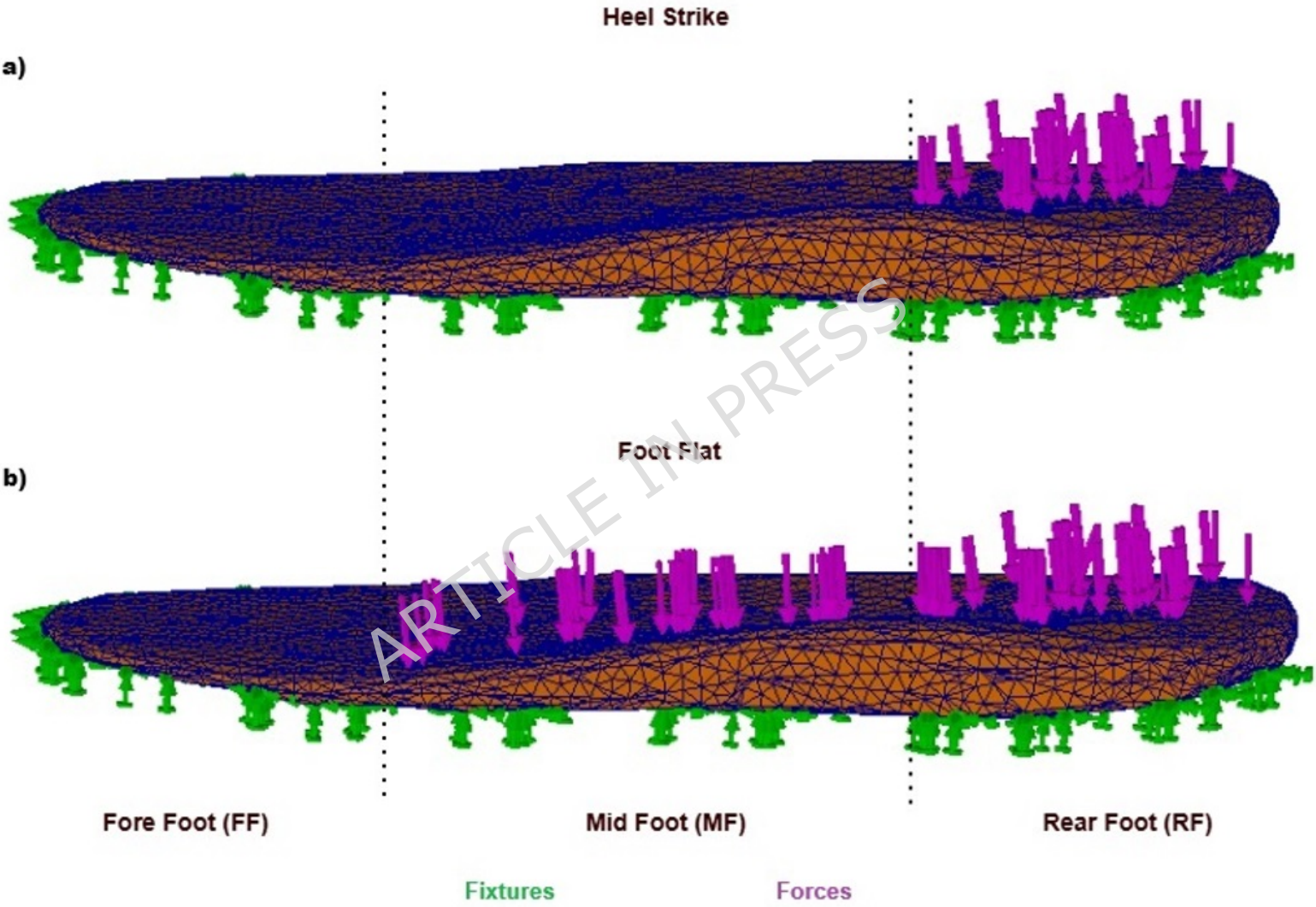

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Table 1. Finite Element Analysis model parameters and settings

Parameter	Value	Description
Software	SolidWorks 2021 (Dassault Systèmes)	Used for model creation, meshing, and static structural analysis.
Material Model	Linear elastic, isotropic	Appropriate for comparative static analysis of EVA foam under immediate loading [42, 43].
Young's Modulus (E)	8.52 ± 1.67 MPa	Derived from tensile testing of the manufactured EVA foam samples.
Poisson's Ratio (ν)	0.4	Typical value for elastomeric foam materials used in orthotics [45].
Density	0.206 ± 0.01 g/cm³	Measured via density testing; used for mass calculation in FEA.
Element Type	Tetrahedral (solid)	Suitable for complex 3D geometries with curved voids.
Mesh Size (global)	~18,886 nodes, 9,442 elements	Determined via mesh convergence analysis (<0.2% change in displacement).
Local Mesh Refinement	Applied around offloading voids	Ensured accurate resolution of high stress gradients.
Boundary Conditions	Bottom surface fully fixed (all DOFs)	Simulates firm contact between insole and ground.
Loading – Heel Strike	15 kg force (~25% body weight) applied to rearfoot	Represents initial contact phase based on gait analysis.
Loading – Foot Flat	19.62 kg rearfoot + 10.37 kg midfoot (total ~50% body weight)	Represents full weight-bearing phase.
Solver Type	Static nonlinear	Used to compute von Mises stress and nodal displacements.

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Table 2. Detailed gait results of off-loading geometries for each foot.

		Region	Barefoot (g/cm²)	Cylinder (g/cm²)	Sphere (g/cm²)	Cone (g/cm²)
LEFT FOOT	Rear Foot	Medial	42.9	45.8	42.3	47.2
		Lateral	45.8	40.3	46.1	41.8
	Mid Foot	Medial	36.3	26.0	26.3	22.9
		Lateral	40.0	39.7	43.3	35.6
	Fore Foot	Medial	37.5	29.8	31.1	30.8
		Lateral	46.6	31.2	37.1	31.9
RIGHT FOOT	Rear Foot	Medial	46.7	39.3	36.3	41.7
		Lateral	44.2	35.0	40.8	45.0
	Mid Foot	Medial	42.5	29.7	30.1	23.6
		Lateral	45.5	45.1	51.8	38.5
	Fore Foot	Medial	33.9	31.4	33.2	29.1
		Lateral	41.0	35.2	31.7	30.5
Off-loading	Volume of a cylinder: $V = \pi r^2 h$ Volume of a section of a sphere: $V = \frac{1}{3}\pi h^2(3r - h)$ Volume of a cone: $V = \frac{1}{3}\pi h^2$			20 - 5 (Diameter, mm - Depth, mm) 1.57 cm³	20 - 5 (Diameter, mm - Depth, mm) 0.654 cm³	20 - 5 (Diameter, mm - Depth, mm) 0.524 cm³
P.max	Max point of pressure, P.max (g/cm²)	LF	1834.0±12.7	1422.9±5.7	1275.6±1.8	1200.1±4.1
		RF	1716.7±12.5	1385.5±4.5	1143.4±5.0	1133.5±3.1

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Table 4. Summary of biomechanical efficacy and the comparison of offloading geometries

Geometry	Gait Analysis: Max Peak Pressure Reduction (vs. Barefoot)	FEA: Internal Stress Concentration (Qualitative)	FEA: Structural Stability/Durability (Qualitative)	Overall Biomechanical Recommendation
Conical	34.5% (Greatest Reduction)	Highest (Localized stress points)	Lowest (Potential for early failure)	Best for maximum offloading, but requires careful material selection.
Spherical	33.4% (High Reduction)	Moderate (Favorable stress distribution)	High (Good durability)	Most balanced approach: adequate offloading with favorable internal stress.
Cylindrical	22.3% (Consistent Reduction)	Lowest (Uniform stress values)	Highest (Predictable, robust performance)	Most consistent and robust, but with slightly lower peak offloading.

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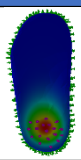
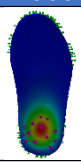
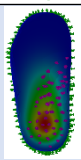
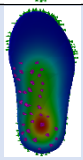
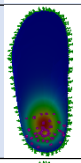
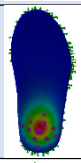
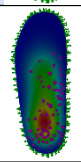
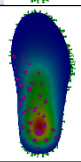
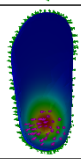
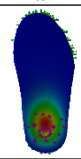
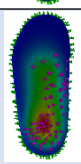
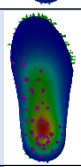
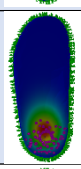
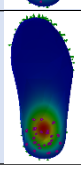
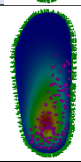
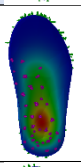
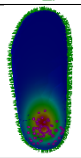
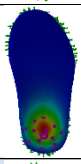
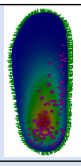
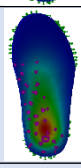
Table 3. FEA results of different off-loading geometries in terms of vonMises stress and material displacement values at each gait phase.

Off-loading Type	Gait Phase	Forces (kgF)	Results (max)	Left Foot	Right Foot	Simulation Results (*URES)	
						Left Foot	Right Foot
Without any type	Heel Strike, RearF (25%)	RearF: 15 MidF: 0 ForeF: 0	vonMises (MPa)	0.425	0.442		
			Displacement (mm)	1.8	1.48		
	Foot Flat RearF, MidF (50%)	RearF: 19.62 MidF: 10.37 ForeF: 0	vonMises (MPa)	0.601	0.597		
			Displacement (mm)	2.82	2.07		
Cone (d=10mm)	Heel Strike, RearF (25%)	RearF: 15 MidF: 0 ForeF: 0	vonMises (MPa)	0.37	0.406		
			Displacement (mm)	1.49	1.38		
	Foot Flat RearF, MidF (50%)	RearF: 19.62 MidF: 10.37 ForeF: 0	vonMises (MPa)	0.513	0.586		
			Displacement (mm)	2.22	2.01		
Cone (d=20mm)	Heel Strike, RearF (25%)	RearF: 15 MidF: 0 ForeF: 0	vonMises (MPa)	0.353	0.429		
			Displacement (mm)	1.61	1.46		
	Foot Flat RearF, MidF (50%)	RearF: 19.62 MidF: 10.37 ForeF: 0	vonMises (MPa)	0.481	0.594		
			Displacement (mm)	2.32	2.16		
Cone (d=30mm)	Heel Strike, RearF (25%)	RearF: 15 MidF: 0 ForeF: 0	vonMises (MPa)	0.574	0.378		
			Displacement (mm)	1.38	1.33		
	Foot Flat RearF, MidF (50%)	RearF: 19.62 MidF: 10.37 ForeF: 0	vonMises (MPa)	0.761	0.542		
			Displacement (mm)	2.1	2.09		
Sphere (d=10mm)	Heel Strike, RearF (25%)	RearF: 15 MidF: 0 ForeF: 0	vonMises (MPa)	0.388	0.412		
			Displacement (mm)	1.58	1.37		
	Foot Flat RearF, MidF (50%)	RearF: 19.62 MidF: 10.37 ForeF: 0	vonMises (MPa)	0.535	0.595		
			Displacement (mm)	2.33	2.02		
Sphere (d=20mm)	Heel Strike,	RearF: 15 MidF: 0	vonMises (MPa)	0.369	0.376		

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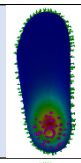
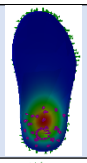
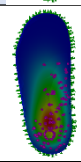
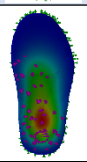
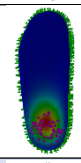
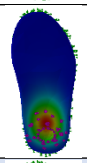
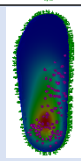
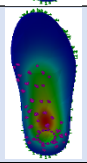
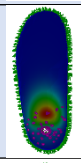
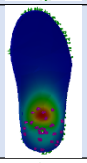
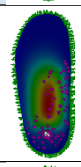
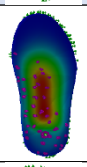
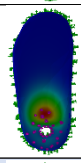
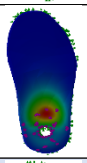
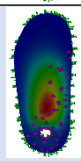
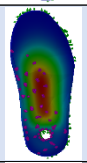
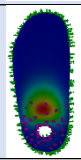
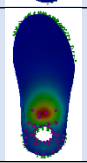
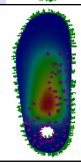
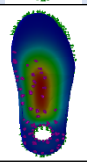
	RearF (25%)	ForeF: 0	Displacement (mm)	1.69	1.36		
	Foot Flat RearF, MidF (50%)	RearF: 19.62 MidF: 10.37 ForeF: 0	vonMises (MPa)	0.528	0.525		
			Displacement (mm)	2.61	2.07		
Sphere (d=30mm)	Heel Strike, RearF (25%)	RearF: 15 MidF: 0 ForeF: 0	vonMises (MPa)	0.651	0.382		
			Displacement (mm)	1.54	1.4		
	Foot Flat RearF, MidF (50%)	RearF: 19.62 MidF: 10.37 ForeF: 0	vonMises (MPa)	0.986	0.536		
			Displacement (mm)	2.33	2.1		
Cylinder (d=10mm)	Heel Strike, RearF (25%)	RearF: 15 MidF: 0 ForeF: 0	vonMises (MPa)	0.405	0.384		
			Displacement (mm)	0.609	0.634		
	Foot Flat RearF, MidF (50%)	RearF: 19.62 MidF: 10.37 ForeF: 0	vonMises (MPa)	0.635	0.579		
			Displacement (mm)	1.29	1.21		
Cylinder (d=20mm)	Heel Strike, RearF (25%)	RearF: 15 MidF: 0 ForeF: 0	vonMises (MPa)	0.362	0.32		
			Displacement (mm)	0.549	0.552		
	Foot Flat RearF, MidF (50%)	RearF: 19.62 MidF: 10.37 ForeF: 0	vonMises (MPa)	0.63	0.501		
			Displacement (mm)	1.29	1.15		
Cylinder (d=30mm)	Heel Strike, RearF (25%)	RearF: 15 MidF: 0 ForeF: 0	vonMises (MPa)	0.324	0.247		
			Displacement (mm)	0.714	0.478		
	Foot Flat RearF, MidF (50%)	RearF: 19.62 MidF: 10.37 ForeF: 0	vonMises (MPa)	0.497	0.42		
			Displacement (mm)	1.39	1.17		

*URES: Resultant displacement.
**d=diameter.

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